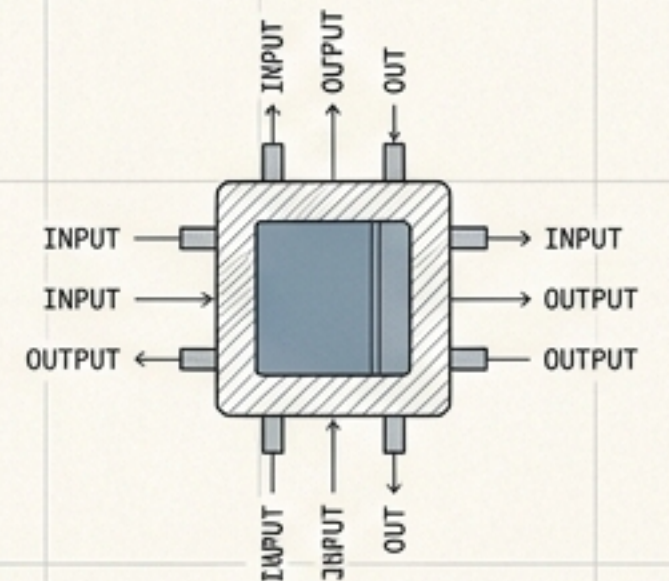
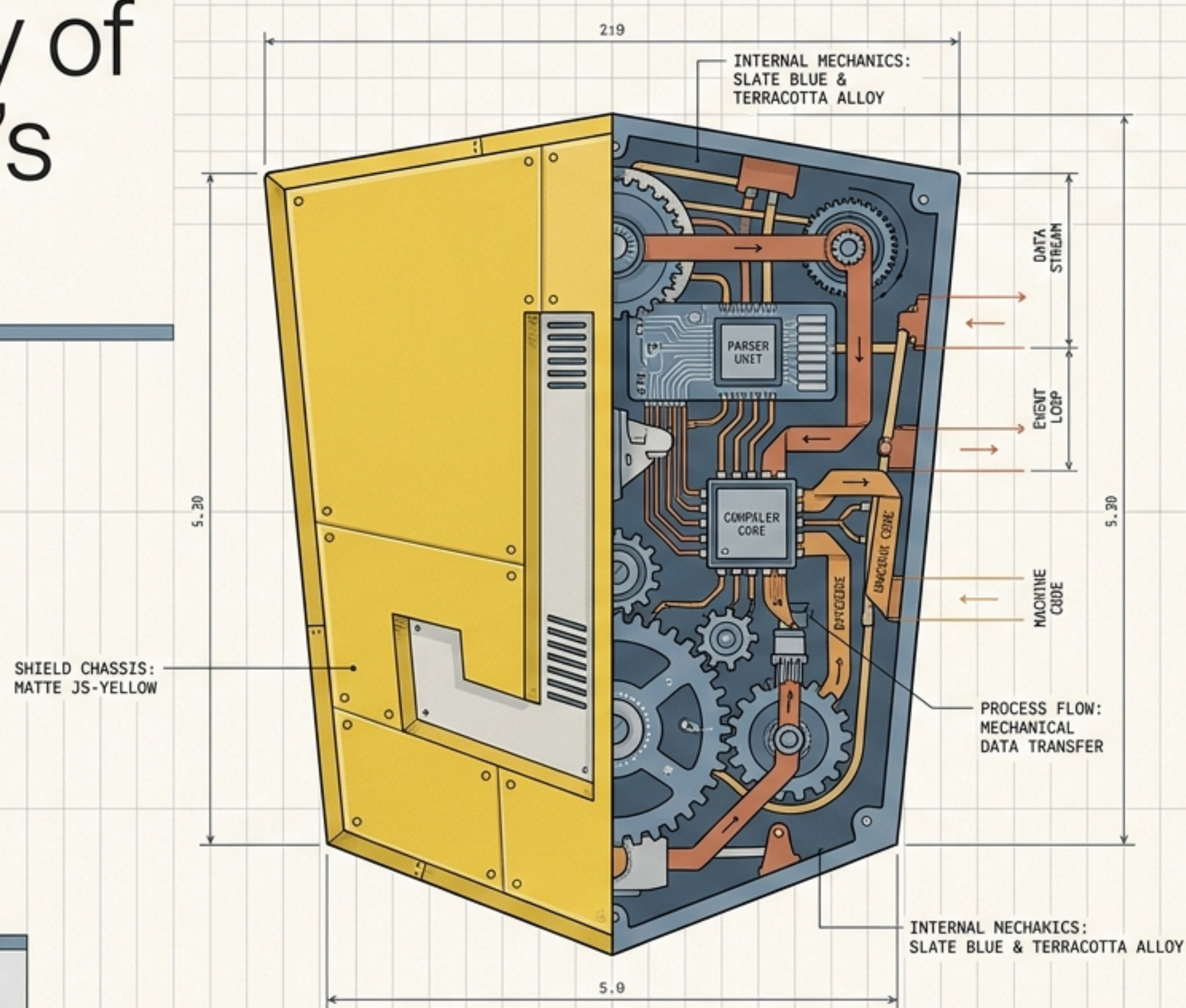
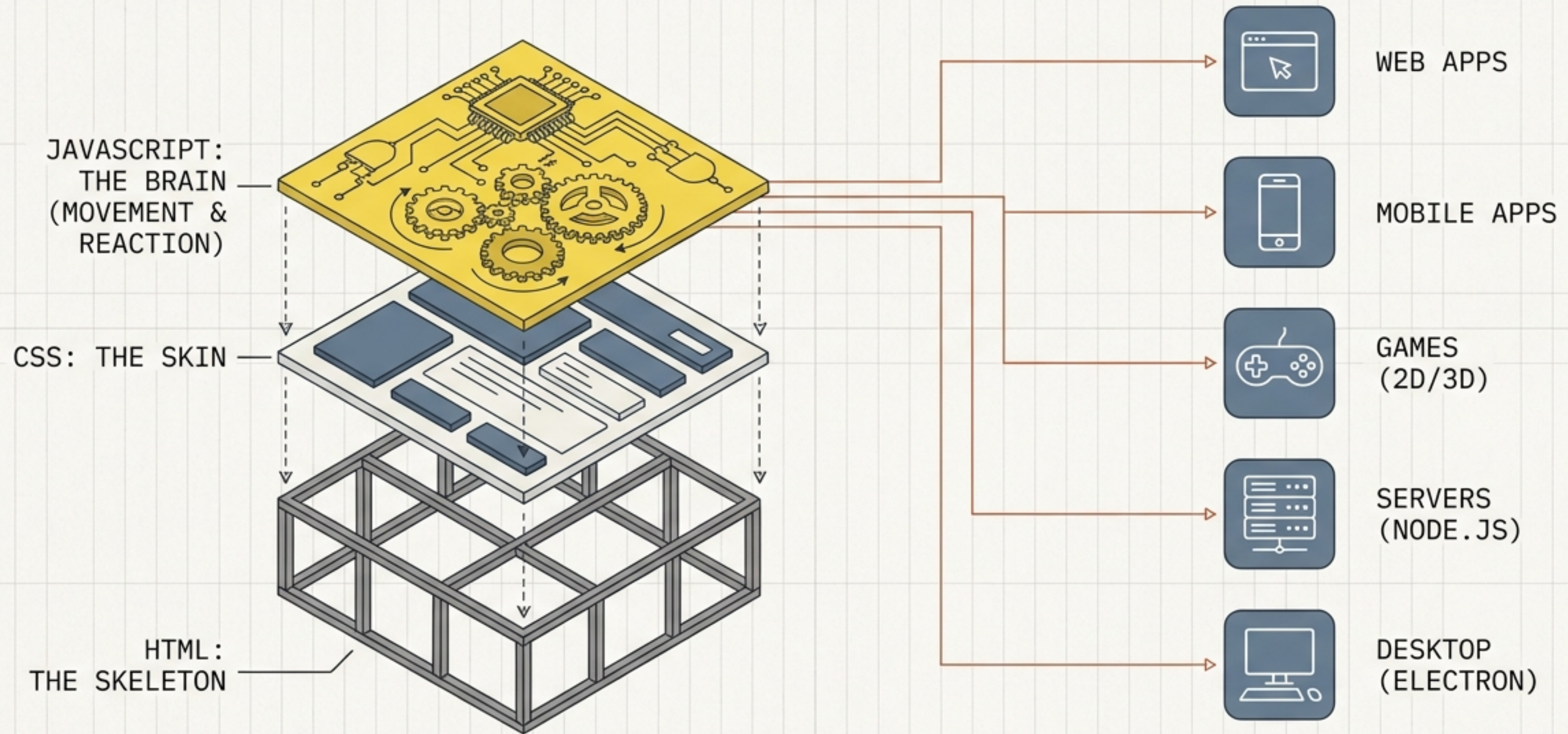


Anatomy of the Web's Brain

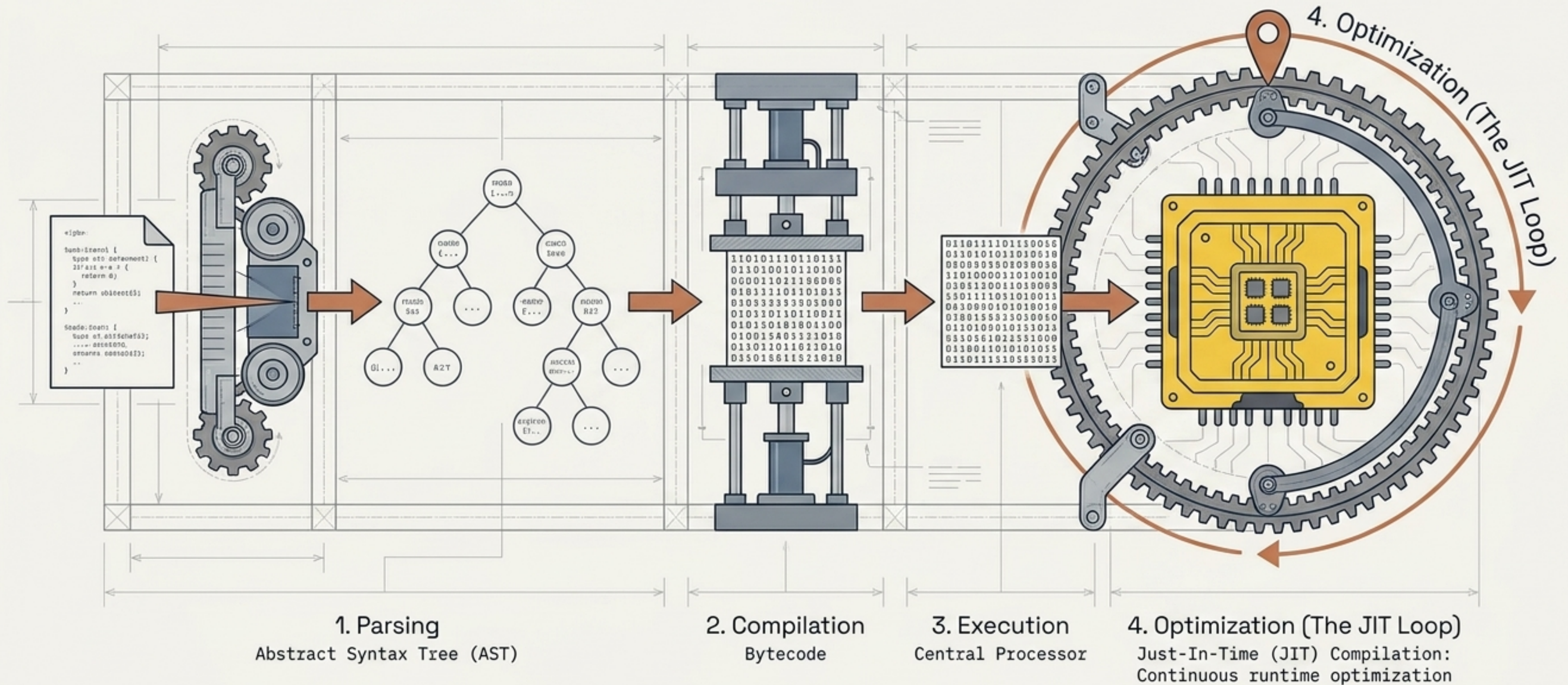
```
> Execute teardown sequence:  
js-engine --verbose
```



THE TRINITY AND THE ECOSYSTEM



The Processing Pipeline



Constructing the Execution Environment

Function Execution Context (FEC)

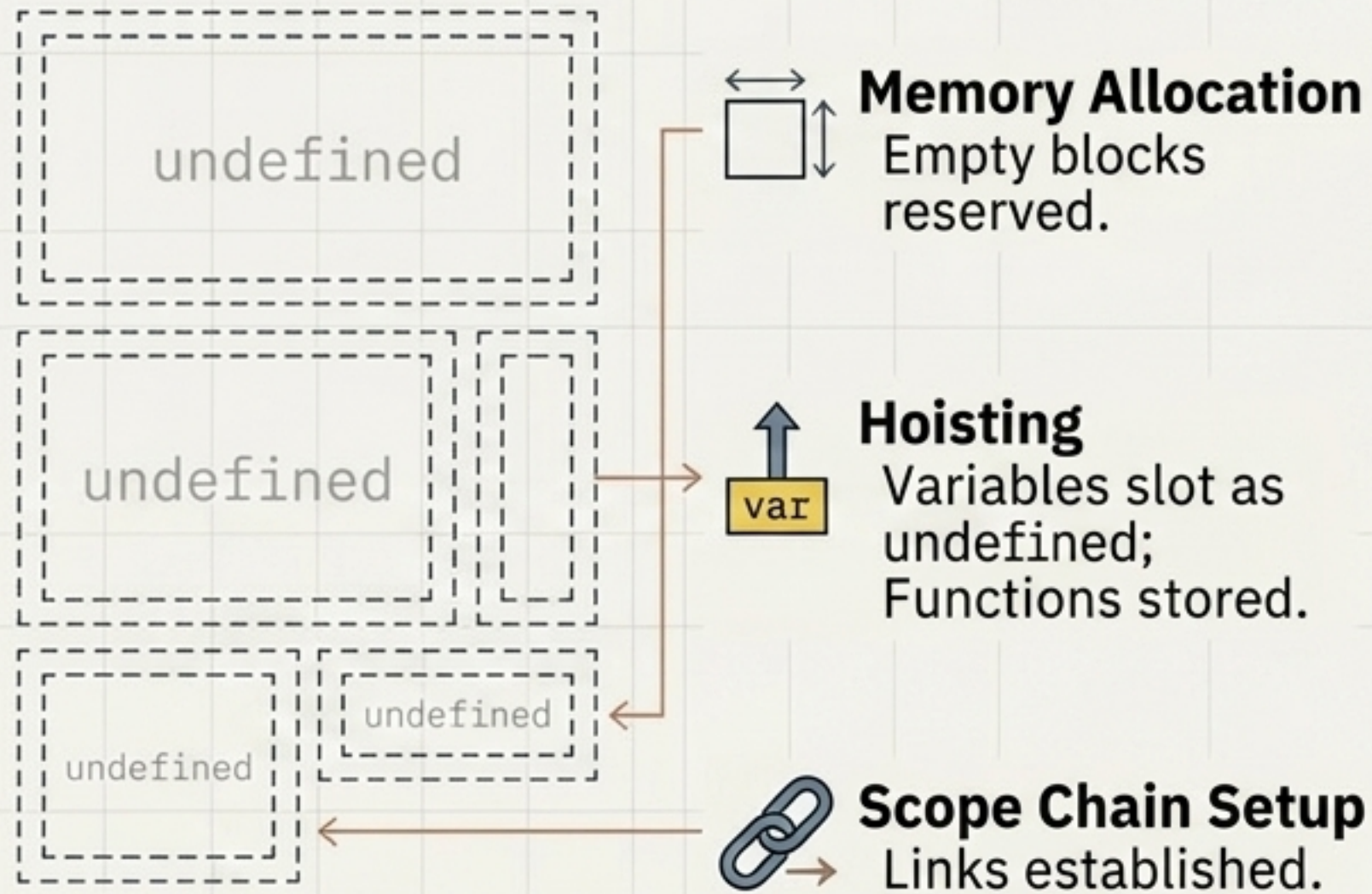
- Created dynamically each time a function is called.
- Temporary and isolated.
- Multiple can exist simultaneously.

Global Execution Context (GEC)

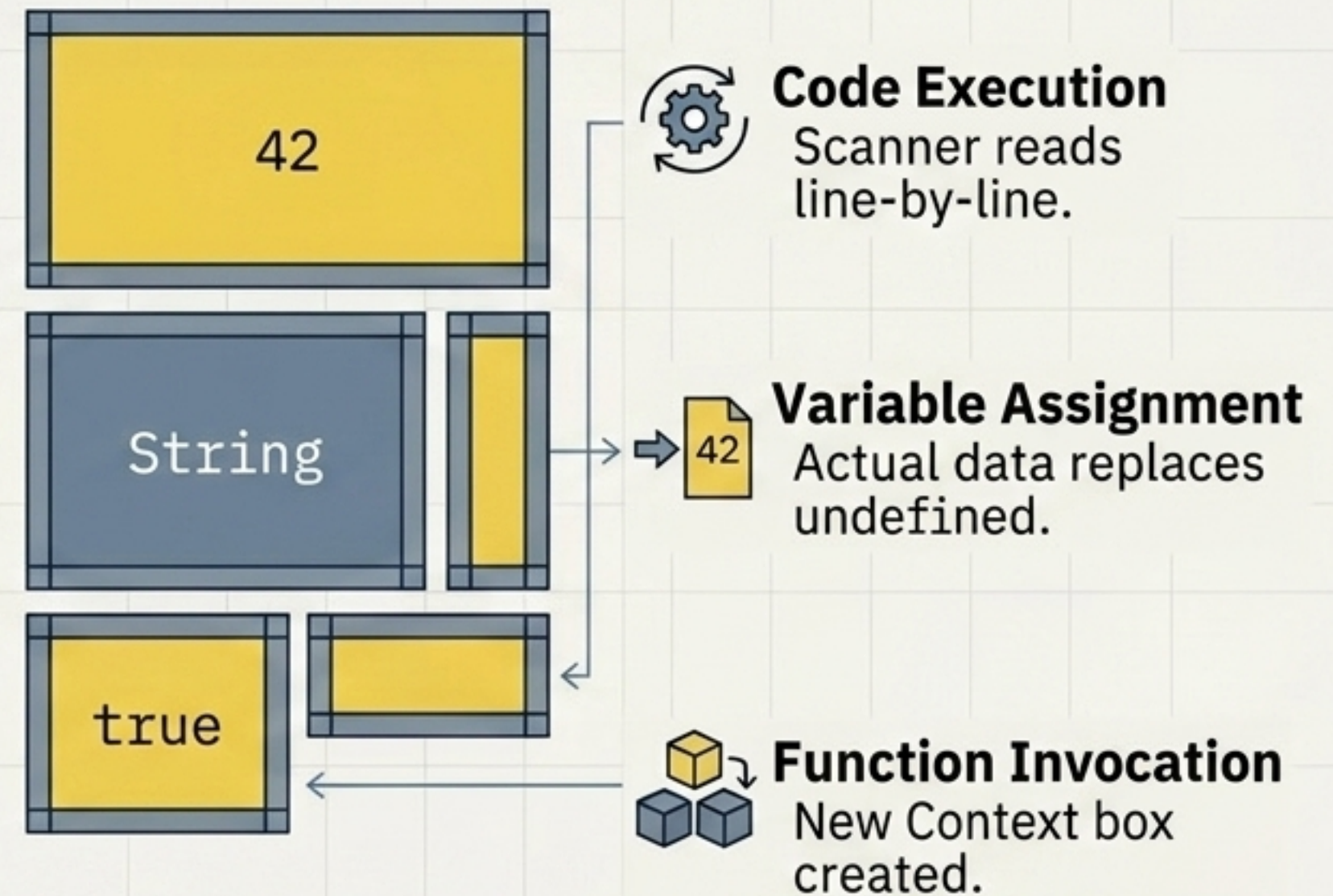
- Created automatically for code outside any function.
- The foundational base layer.
- Only one exists per script.

The Two-Phase Context Lifecycle

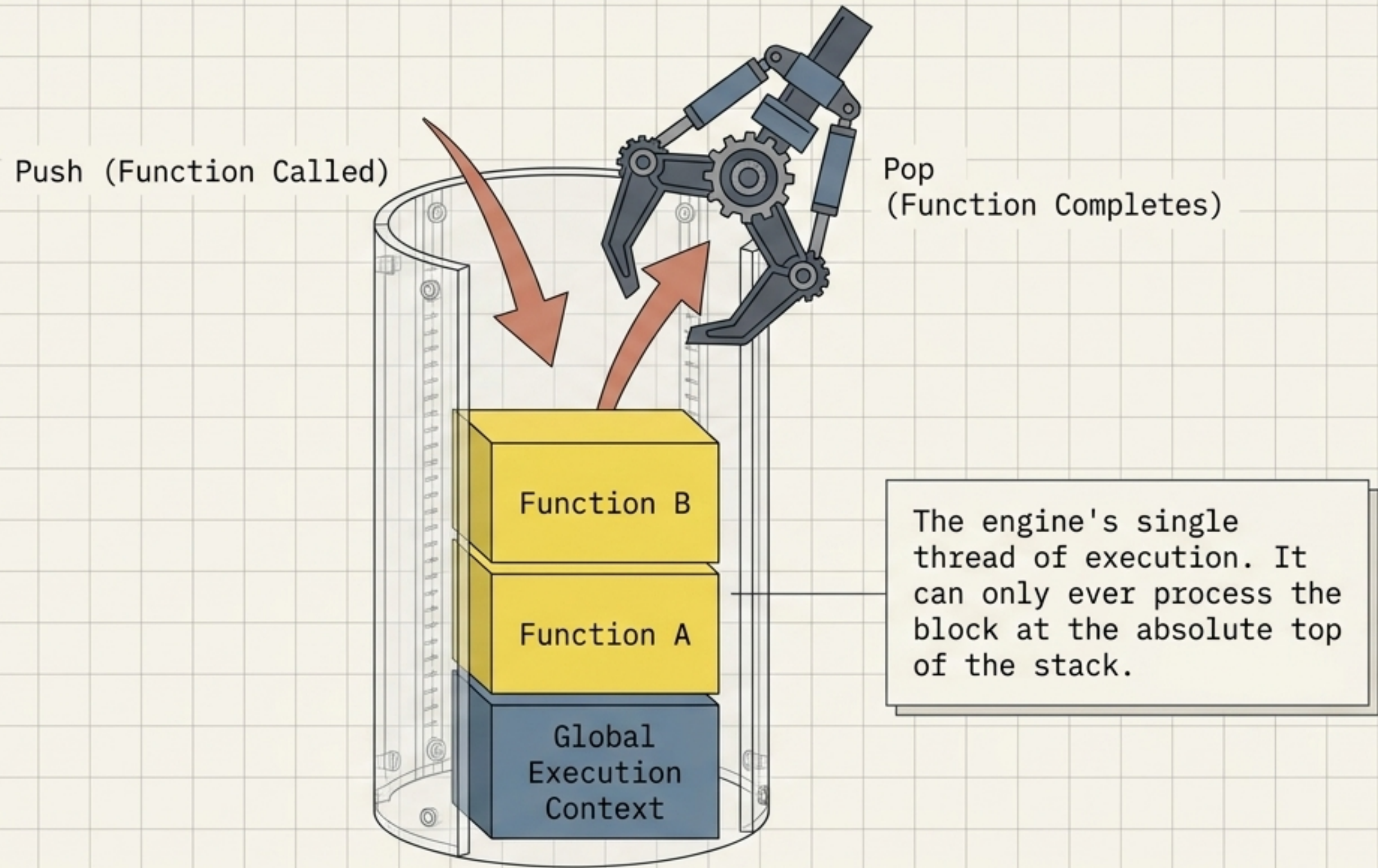
Phase 1: Creation (Setup)



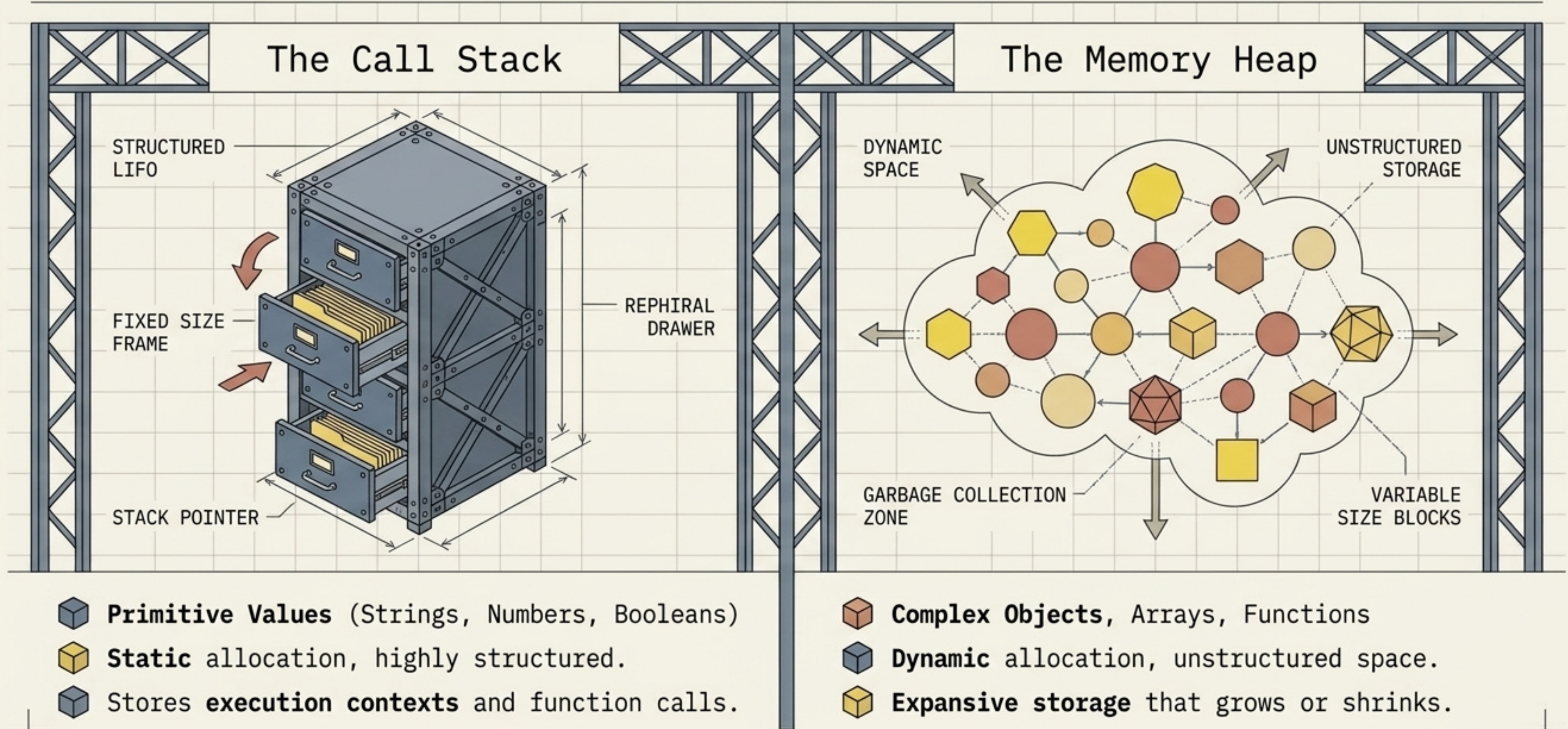
Phase 2: Execution (Action)



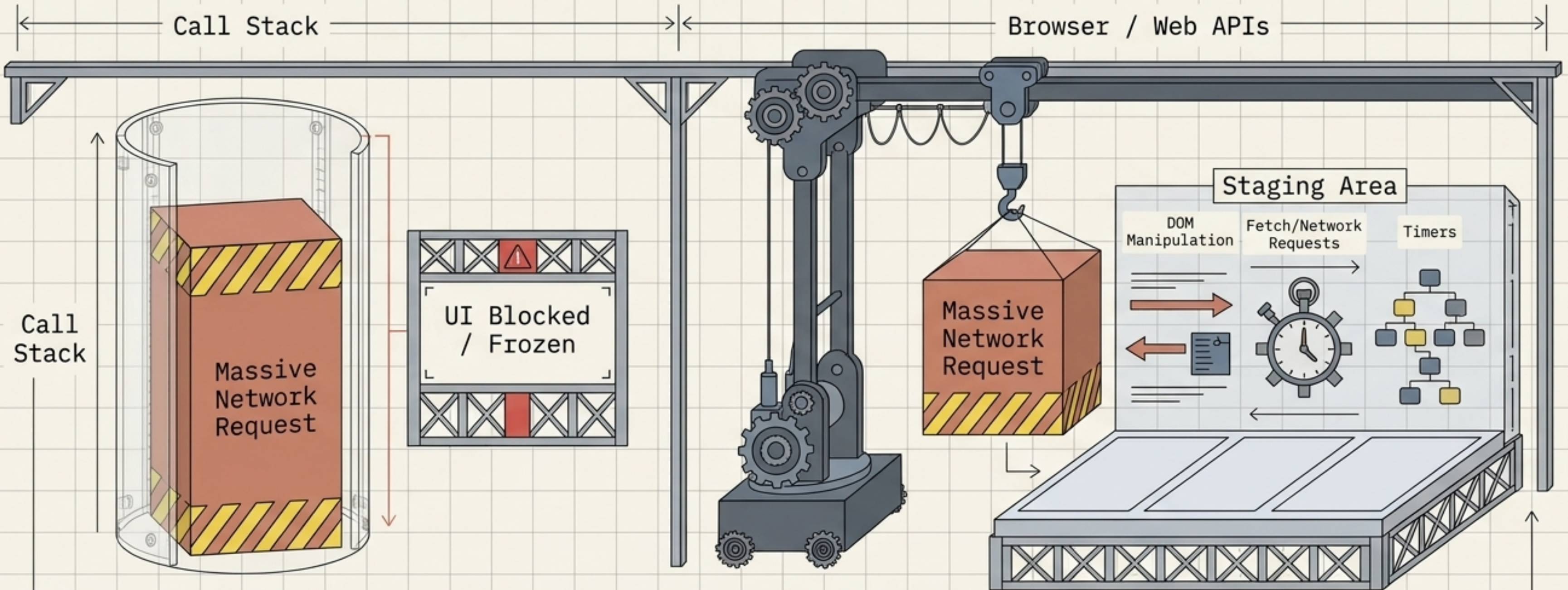
Orchestrating Order: The Call Stack



Memory Allocation Matrix



The Single-Thread Paradox



JavaScript itself is strictly single-threaded. To multitask, it relies entirely on the external host environment to do the heavy lifting in the background.

The Master Schematic

